Important Long-Answer Questions (Revised – 5 Marks Each)

1. What are web browsers and web servers? Describe their working with examples.
2. Differentiate between **client-side scripting** and **server-side scripting** with examples.
3. Write short notes on **web applications** and their importance in daily life.
4. Explain the structure of an HTML document with proper syntax and example code.
5. What are **HTML lists**? Explain ordered, unordered, and definition lists with examples.
6. Discuss the different ways to **embed CSS in HTML** with examples.
7. Explain the concept of **linking documents in HTML**. Write an example of hyperlinking.
8. Differentiate between HTML, DHTML, and CSS. How do they enhance web page design?
9. What is an IP Address? Explain its classes with examples.

Short Questions(1Mark)

1. What is the Internet?
2. Define the World Wide Web (WWW).
3. What is the difference between Internet and Intranet?
4. Define ISP. Give one example.
5. What is an IP address?
6. What is a URL? Give one example.
7. Write the full form of HTTP and DNS.
8. What is the function of a web browser? Name two browsers.
9. Define a web server.
10. What are web applications? Give two examples.
11. Name two tools used for website creation.
12. What is HTML?
13. Write the difference between ordered and unordered lists in HTML.
14. What is the use of `<img>` tag in HTML?
15. How do you create a hyperlink in HTML?
16. What is the difference between client-side scripting and server-side scripting?
17. What is the use of `<table>` tag in HTML?
18. What is DHTML?
19. Write two advantages of using CSS.
20. Name three ways to embed CSS in an HTML page.

Answers:

Long Questions

1. What are web browsers and web servers? Describe their working with examples.

A **web browser** is a software application that allows users to access and view websites on the Internet. Popular browsers include **Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge**. A browser interprets HTML, CSS, and JavaScript code written on web pages and presents it in a user-friendly graphical interface. When a user types a URL, the browser sends an HTTP request to a web server, retrieves the requested page, and displays it.

A **web server**, on the other hand, is software or hardware that stores and delivers web pages to clients (browsers) over the Internet. Popular web servers are **Apache, Nginx, and Microsoft IIS**. Web servers handle requests sent via HTTP or HTTPS, process them, and return responses in the form of HTML documents, images, or other resources.

Working process:

1. A user enters a URL in the browser.
2. The browser translates the domain name into an IP address using DNS.
3. A request is sent to the appropriate web server.
4. The server locates the requested file (e.g., index.html).
5. The server sends the file back to the browser.
6. The browser renders the file and displays the web page to the user.

Example: If you enter `https://www.wikipedia.org`, the browser sends a request to the Wikipedia server. The server processes it and sends back the HTML, CSS, and images of the homepage, which the browser displays.

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2. Differentiate between client-side scripting and server-side scripting with examples.

Client-side scripting refers to scripts that run directly on the user's browser. It mainly uses **JavaScript, HTML, and CSS** to make web pages interactive. For example, form validation, dropdown menus, and animations are implemented on the client-side. The advantage is that it reduces load on the server, but it can be manipulated by users, which may affect security.

Server-side scripting runs on the server before the page is sent to the client. It uses languages such as **PHP, Python, ASP.NET, Node.js, and Java**. It is responsible for database interaction, user authentication, and dynamic page generation. The advantage is that it is secure and powerful, but it may increase server load.

Key Differences:

- Client-side scripting executes in the browser, while server-side scripting executes on the server.
- Client-side uses JavaScript, while server-side uses PHP/ASP/Python.
- Client-side is mainly for presentation, while server-side manages logic and data.

Example:

- Client-side: JavaScript checking if an email field is empty before form submission.
- Server-side: PHP script verifying login credentials from a database.

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3. Write short notes on web applications and their importance in daily life.

A **web application** is a software program that runs on a web server and is accessed through a web browser. Unlike traditional desktop applications, web applications don't require installation; they run online. Examples include **Gmail, Google Docs, Facebook, and Online Banking Systems**.

Importance in daily life:

1. **Accessibility:** Web apps can be accessed from anywhere using a browser.
2. **Ease of updates:** They are updated centrally on the server, so users don't need to install updates.
3. **Cross-platform support:** They work on different devices (mobile, desktop, tablets).
4. **Productivity:** Tools like Google Docs and MS Office 365 improve collaboration.
5. **E-commerce & services:** Amazon, Flipkart, and banking apps allow shopping and transactions easily.

Example: Online banking applications enable users to transfer money, check balances, and pay bills without visiting the bank physically. Similarly, e-learning web apps like Coursera and Zoom support remote education.

Thus, web applications have become an essential part of our personal and professional lives, making tasks more efficient and convenient.

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4. Explain the structure of an HTML document with proper syntax and example code.

An **HTML document** is the backbone of any web page. It defines the structure, content, and elements of a page using tags. Every HTML page follows a particular structure consisting of **doctype, head, and body sections**.

Basic Structure:

```
<!DOCTYPE html>
<html>
<head>
   <title>My First Web Page</title>
</head>
<body>
   <h1>Welcome to HTML</h1>
   <p>This is a sample paragraph.</p>
</body>
</html>
```

Explanation:

- `<!DOCTYPE html>`: Defines the type of document and version of HTML (HTML5 here).
- `<html>`: Root element that contains all HTML code.
- `<head>`: Contains metadata, title, and links to CSS/JS files.
- `<title>`: Defines the title of the webpage shown in the browser tab.
- `<body>`: Contains the main content like text, images, tables, and links.

The structure ensures that browsers interpret and render the page correctly.

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5. What are HTML lists? Explain ordered, unordered, and definition lists with examples.

HTML lists allow us to display information in a structured format. There are three main types of lists:

1. **Ordered List (`<ol>`)**
   Displays items in a sequence using numbers or letters.

```
<ol>
   <li>HTML</li>
   <li>CSS</li>
   <li>JavaScript</li>
</ol>
```

Output: 1. HTML 2. CSS 3. JavaScript

2. **Unordered List (`<ul>`)**
   Displays items with bullet points.

```
<ul>
   <li>Apples</li>
   <li>Bananas</li>
   <li>Oranges</li>
</ul>
```

Output: ● Apples ● Bananas ● Oranges

3. **Definition List (`<dl>`)**
   Displays terms with descriptions.

```
<dl>
   <dt>HTML</dt>
   <dd>Hyper Text Markup Language</dd>
   <dt>CSS</dt>
   <dd>Cascading Style Sheets</dd>
</dl>
```

Lists are widely used in menus, navigation bars, and structured data presentation.

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6. Discuss the different ways to embed CSS in HTML with examples.

CSS (Cascading Style Sheets) defines the design and layout of web pages. There are **three ways to embed CSS in HTML**:

1. **Inline CSS**
   Applied directly to an element using the `style` attribute.

```
<p style="color:red;">This is red text.</p>
```

2. **Internal CSS**
   Defined inside `<style>` tags in the `<head>` section.

```
<head>
<style>
p { color: blue; font-size: 18px; }
</style>
</head>
```

3. **External CSS**
   Stored in a separate `.css` file and linked using `<link>`.

```
<link rel="stylesheet" type="text/css" href="style.css">
```

Comparison:

- Inline CSS is quick but not reusable.
- Internal CSS is good for single pages.
- External CSS is the most efficient for large websites, ensuring consistency.

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7. Explain the concept of linking documents in HTML. Write an example of hyperlinking.

Linking documents allows users to navigate between different pages or sections of a website. In HTML, links are created using the `<a>` (anchor) tag. Hyperlinks may connect to other pages, documents, emails, or sections of the same page.

Example:

```
<a href="https://www.google.com">Visit Google</a>
```

This creates a clickable link that redirects users to Google.

Types of Linking:

1. **External linking:** Links to another website.
2. **Internal linking:** Links to another page within the same site.
3. **Anchor linking:** Jumps to a specific section within the same page using `id`.

Hyperlinking is one of the most powerful features of the web, creating a vast interconnected system of documents known as the World Wide Web.

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8. Differentiate between HTML, DHTML, and CSS. How do they enhance web page design?

HTML (HyperText Markup Language):

- Standard language for structuring web pages.
- Defines headings, paragraphs, images, tables, and links.
- Example: `<h1>Hello World</h1>`.

DHTML (Dynamic HTML):

- Extension of HTML with **CSS, JavaScript, and DOM**.
- Allows dynamic content updates without reloading.
- Used for animations, interactive menus, and real-time changes.

CSS (Cascading Style Sheets):

- Defines the style, layout, and appearance of web pages.
- Controls colors, fonts, spacing, and positioning.
- Example: `h1 {color: blue;}`.

Comparison:

- HTML builds the structure.
- CSS designs the look.
- DHTML makes it interactive.

Enhancement in Web Design:

- HTML provides foundation.
- CSS ensures consistent styling.
- DHTML improves user experience with real-time interactivity.

9. What is an IP Address? Explain its classes with examples.

An **IP Address (Internet Protocol Address)** is a unique numerical identifier assigned to each device connected to a network. It helps in identifying and locating devices so that data can be sent and received correctly over the Internet. Every device like a computer, mobile phone, or printer must have an IP address to communicate.

There are two main versions: **IPv4 (32-bit)** and **IPv6 (128-bit)**. IPv4 is written as four decimal numbers separated by dots (e.g., 192.168.1.1), while IPv6 uses hexadecimal numbers separated by colons (e.g., 2001:0db8::1).

Classes of IPv4 Addresses

IPv4 addresses are divided into **five classes (A–E)** based on their leading bits and range.

1. **Class A**

- Range: 1.0.0.0 to 126.0.0.0
- Default Subnet Mask: 255.0.0.0
- Supports **16 million hosts** on each of 128 networks.
- Example: 10.1.2.3

2. **Class B**

- Range: 128.0.0.0 to 191.255.0.0
- Default Subnet Mask: 255.255.0.0

- Supports **65,000 hosts** on each of 16,000 networks.
- Example: 172.16.5.4

3. **Class C**

- Range: 192.0.0.0 to 223.255.255.0
- Default Subnet Mask: 255.255.255.0
- Supports **254 hosts** on each of 2 million networks.
- Example: 192.168.1.1 (commonly used in home networks).

4. **Class D** (Multicast)

- Range: 224.0.0.0 to 239.255.255.255
- Used for multicast groups, not for normal host addressing.
- Example: 224.0.0.1

5. **Class E** (Experimental)

- Range: 240.0.0.0 to 255.255.255.255
- Reserved for research and future use.
- Example: 250.1.1.1

Special Addresses

- **127.0.0.1** → Loopback address (used for testing).
- **Private IP ranges** → Class A (10.x.x.x), Class B (172.16.x.x – 172.31.x.x), Class C (192.168.x.x).
- **Public IP** → Assigned by ISPs, used on the Internet.

📌 Short Questions with Answers

1. What is the Internet?
The Internet is a global network of interconnected computers that communicate through standardized protocols. It allows sharing of data, emails, multimedia, and resources worldwide. It is often called the “network of networks.”

2. Define the World Wide Web (WWW).
The World Wide Web (WWW) is a collection of information and resources accessed through the Internet. It uses web pages linked together by hyperlinks and accessed via browsers. It works mainly on the HTTP protocol.

3. What is the difference between Internet and Intranet?
The Internet is a public global network accessible to everyone, whereas an Intranet is a private network used within an organization. The Internet provides worldwide communication, while Intranet restricts access to internal users only.

4. Define ISP. Give one example.
ISP stands for Internet Service Provider, a company that provides Internet connectivity to users. Examples include Airtel, Jio, and BSNL. They offer services like broadband, mobile data, and leased lines.

5. What is an IP address?
An IP address is a unique numerical label assigned to each device connected to a network. It identifies the device’s location on the Internet. Example: 192.168.1.1 (IPv4).

6. What is a URL? Give one example.
URL stands for Uniform Resource Locator, which specifies the address of a resource on the web. Example: `https://www.google.com`. It includes protocol, domain, and path.

7. Write the full form of HTTP and DNS.
HTTP: Hyper Text Transfer Protocol – it is used to transfer data on the web.
DNS: Domain Name System – it translates domain names into IP addresses for easy access.

8. What is the function of a web browser? Name two browsers.
A web browser is a software application used to access and display web pages. Examples: Google Chrome and Mozilla Firefox. It interprets HTML, CSS, and scripts for user interaction.

9. Define a web server with an example.
A web server is software or hardware that delivers web content (pages, images, etc.) to clients. Example: Apache and Microsoft IIS. It responds to HTTP requests from browsers.

10. What are web applications? Give two examples.
Web applications are software programs that run on web browsers over the Internet. Examples include Gmail and Google Docs. They are accessible without installing on the local system.

11. Name two tools used for website creation.
Common website creation tools are WordPress, Wix, and Dreamweaver. These tools help in designing, coding, and hosting websites easily, often with drag-and-drop features.

12. What is HTML?
HTML stands for Hyper Text Markup Language. It is the standard language for creating web pages. HTML defines the structure of content using tags.

13. Write the difference between ordered and unordered lists in HTML.
Ordered lists (`<ol>`) display items with numbers or letters in sequence. Unordered lists (`<ul>`) display items with bullet points. Both use the `<li>` tag for list items.

14. What is the use of `<img>` tag in HTML?
The `<img>` tag is used to insert images into a webpage. It requires attributes like `src` (path of image) and `alt` (alternate text). Example: `<img src="photo.jpg" alt="My Photo">`.

15. How do you create a hyperlink in HTML?
A hyperlink is created using the `<a>` tag. Example: `<a href="https://www.google.com">Google</a>`. It connects one webpage to another or a resource.

16. What is the difference between client-side scripting and server-side scripting?
Client-side scripting runs on the user's browser (e.g., JavaScript), making pages interactive. Server-side scripting runs on the server (e.g., PHP), generating dynamic content before sending it to the client.

17. What is the use of `<table>` tag in HTML?
The `<table>` tag is used to display data in rows and columns. It uses `<tr>` for rows, `<td>` for data cells, and `<th>` for headings. Tables organize structured data.

18. What is DHTML?
DHTML stands for Dynamic HTML. It combines HTML, CSS, and JavaScript to create interactive and animated web pages. It allows real-time changes without reloading the page.

19. Write two advantages of using CSS.
CSS separates content from design, making pages easier to maintain. It allows consistent styling across multiple pages and reduces coding repetition. It also improves page loading speed.

20. Name three ways to embed CSS in an HTML page.

1. Inline CSS (using `style` attribute inside tags).
2. Internal CSS (inside `<style>` tag in `<head>`).
3. External CSS (linking `.css` file using `<link>`).